

COS30018 – Intelligent Systems

Week 6 tutorial

Naïve Bayes (NB) with sklearn

Naive Bayes is the most straightforward and fast classification algorithm based on Bayes' Theorem. Naive Bayes classifier is used in various applications such as spam filtering, text classification, sentiment analysis, and recommender systems (of course just in the past, now we have more powerful algorithms LOL).

The Bayes's theorem says:

$$P(c|x) = \frac{P(x|c) \cdot P(c)}{P(x)}$$

Where:

- $P(c|x)$: is the posterior probability of class (c, target) given predictor (x, attributes).
- $P(x|c)$: is the probability of predictor given class.
- $P(c)$: is the prior probability of class.
- $P(x)$: is the prior probability of predictor.

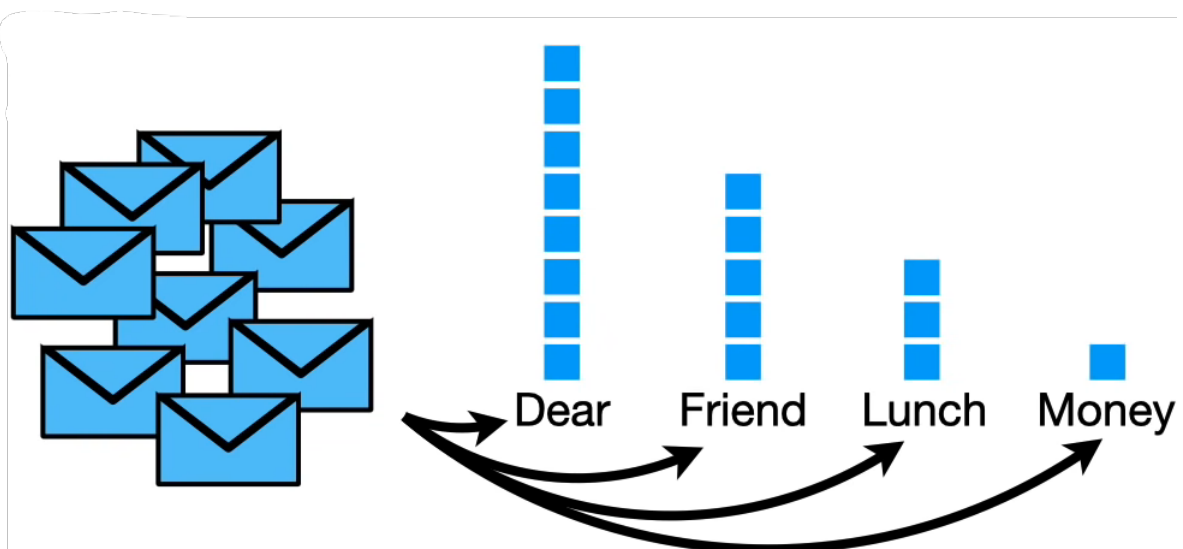
We will understand more about the meaning of each component when we look at an example.

Let's go straight to an example of how NB works as a supervised classification algorithm.

Scenario: We work for Gmail and our task is to classify if an email is spam or not.

Let's say in our dataset, we have **12 emails**: 8 normal emails and 4 spam emails.

The first thing to do is to count the occurrences of all words in the normal emails. Let's say there are only 4 words in the 8 normal emails:



Then calculate the probabilities of seeing each word, given that it was in normal email:

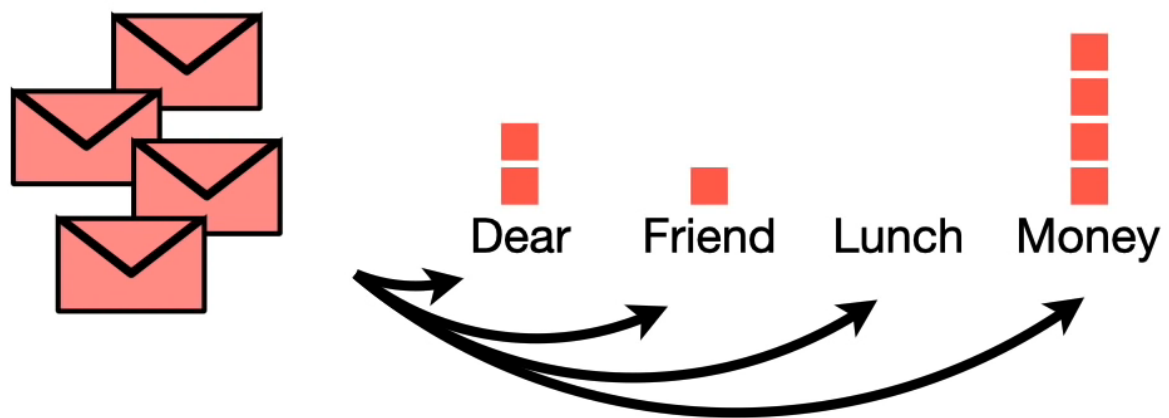
$$p(\text{Dear}|N) = \frac{8}{17} = 0.47$$

$$p(\text{Friend}|N) = \frac{5}{17} = 0.29$$

$$p(\text{Lunch}|N) = \frac{3}{17} = 0.18$$

$$p(\text{Money}|N) = \frac{1}{17} = 0.06$$

We do the same thing for spam emails:



$$p(\text{Dear}|S) = \frac{2}{7} = 0.29$$

$$p(\text{Friend}|S) = \frac{1}{7} = 0.14$$

$$p(\text{Lunch}|S) = \frac{0}{7} = 0$$

$$p(\text{Money}|S) = \frac{4}{7} = 0.57$$

Now, imagine if we got a new message that said “Dear Friend”. How can we predict if this is a normal email or a spam?

First, we calculate the score that “Dear Friend” gets if it is a normal email:

$$\text{score}(N) = p(N) \cdot p(\text{Dear}|N) \cdot p(\text{Friend}|N) = \frac{8}{12} \cdot \frac{8}{17} \cdot \frac{5}{17} = 0.09$$

This score is proportional to the probability that the message is normal, given that it says “Dear Friend”:

$$\text{score}(N) \sim p(N|\text{Dear Friend})$$

We do the same to find the score that “Dear Friend” gets if it is a spam email:

$$\text{score}(S) = p(S) \cdot p(\text{Dear}|S) \cdot p(\text{Friend}|S) = \frac{4}{12} \cdot \frac{2}{7} \cdot \frac{1}{7} = 0.01$$

Again, this score is proportional to the probability that the message is spam, given that it says “Dear Friend”:

$$\text{score}(S) \sim p(S|\text{Dear Friend})$$

And because $\text{score}(N) = 0.09 > \text{score}(S) = 0.01$, we can predict that “Dear Friend” is a normal email.

Implementation:

The implementation of Naïve Bayes classifier and explanations for each step are given in the notebook “Naive_Bayes.ipynb”. In this notebook, we use the library Scikit-learn to create and train the classifier.

References:

- StatQuest: Naive Bayes, Clearly Explained!!!
<https://www.youtube.com/watch?v=O2L2Uv9pdDA>